BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, RAWALPINDI

INTERMEDIATE (11TH CLASS) EXAMINATION, 2021

CHEMISTRY

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

Chemistry (Objective Type) — Inter. (Part-I)-A-2021 (For all sessions) — Paper Code: 6487

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 6487

Note: NOTE: Write answers to the questions on objective answer sheet provided. Four possible answers A,B,C & D to each question are given. Which answer you consider correct, fill the corresponding circle A,B,C or D given in front of each question with Marker or pen ink on the answer sheet provided.

Q1. When water Freezes at 0°C, its density decreases due to:

(A) Cubic structure of ice
(B) Empty spaces present in the structure of ice
(C) Change of bond lengths
(D) Change of bond angle

Q2. Which one of the following is an example of cubic system?

(A) Diamond
(B) Borax
(C) Iodine
(D) Graphite

Q3. Brackett series lie in the region:

(A) U.V
(B) I.R
(C) Visible
(D) X-Ray

Q4. Bohr model of atom is contradicted by:

(A) Plank's quantum theory
(B) dual nature of matter
(C) Heisenberg's uncertainity [sic] principle
(D) Newton theory

Q5. The number of bonds in nitrogen molecule is:

(A) one σ and one π
(B) one σ and two π
(C) three σ (sigma) only
(D) two σ and one π

Q6. The covalent radius of Cl-atom is:

(A) 99.4 pm
(B) 80 pm
(C) 70 pm
(D) 66.4 pm

Q7. One calorie is equivalent to:

(A) 0.4184J
(B) 4.184J
(C) 41.84J
(D) 418.4J

Q8. pH value of vinegar is:

(A) 1.1
(B) 2.0
(C) 2.8
(D) 3.5

Q9. For which system does the equilibrium constant, Kc has units of (concentration)$^{-1}$?

(A) $N_2 + 3H_2 \rightleftharpoons 2NH_3$
(B) $H_2 + I_2 \rightleftharpoons 2HI$
(C) $2NO_2 \rightleftharpoons N_2O_4$
(D) $2HF \rightleftharpoons H_2 + F_2$

Q10. 18g of glucose is dissolved in 90g of water. The relative lowering of vapour pressure is equal to:

(A) 1/5
(B) 5.1
(C) 1/51
(D) 6.0

Q11. Stronger the oxidizing agent, greater is the:

(A) Odixation [sic] potential
(B) Reduction potential
(C) Redox potential
(D) E.M.F of cell

Q12. If the rate equation of a reaction: $2A+B \longrightarrow$ Products is, rate=$K[A]^2[B]$, and A is present in large excess, then order of reaction is:

(A) 1
(B) 2
(C) 3
(D) 4

Q13. Isotopes differ in:

(A) properties which depend upon mass
(B) arrangement of electrons in orbitals
(C) chemical properties
(D) their behaviour in electromagnetic field

Q14. Number of isotopes of Tin is/are:

(A) one
(B) eleven
(C) fifteen
(D) eighteen

Q15. Solvent extraction method is a particularly useful technique for separation when the product to be separated is:

(A) non-volatile or thermally unstable
(B) volatile or thermally stable
(C) non-volatile or thermally stable
(D) volatile or thermally unstable

Q16. Pressure remaining constant, at which temperature the volume of a gas will become twice of what it is at 0°C.

(A) 546 °C
(B) 200 °C
(C) 546 K
(D) 273 K

Q17. The partial pressure of oxygen in the lungs is:

(A) 100 torr
(B) 116 torr
(C) 150 torr
(D) 159 torr

Chemistry (Essay Type) — Inter. (Part-I)-A-2021 (For all sessions)

Time Allowed: 2:40 Hours | **Maximum Marks:** 68

Q. Section - I — Marks: 68

Q2. Write short answers of any eight parts from the following.

x 8 = 16 2

(i) Why is actual yield less than theoratial [sic] yield?

(ii) Define Fractional crystallization with example.

(iii) Magnesium atom is twice heavier than that of carbon.

(iv) Define (i) Stationary phase (ii) Distribution co-efficient

(v) Give uses of Chromatography.

(vi) Why absolute zero is unattainable?

(vii) What is (i) Isotherm (ii) Partiel [sic] Pressure

(viii) What are the Faulty points of Kinetic theory of gas?

(ix) Give quantitatively statement of Charles law.

(x) Give any two differences between Ideal and Non Ideal solution.

(xi) Colligative properties are obeyed when solute is non-volatile and solution is dilute. Justify it.

(xii) 23 gram sodium and 238 gram Uranium have equal number of atoms.

Q3. Write short answers of any eight parts from the following.

x 8 = 16 2

(i) Distinguish between Isomorphism and polymorphism.

(ii) Differentiate between continuous and line spectrum.

(iii) How does polarizibility [sic] effect the strength of London Forces?

(iv) What are the favourable conditions for ammonia synthesis on Industrial scale?

(v) Why is it necessary to decrease the pressure in a discharge tube?

(vi) Justify with examples that some reactions occur at higher rate and some may occur at moderate rate.

(vii) Why positive rays are called canal rays?

(viii) Why do crystals change their habit?

(ix) How does the buffer solution act?

(x) Radioactive decay is always a First order reaction.

(xi) Define the terms (i) helix (ii) Deby [sic] Forces

(xii) What is electromagnetic spectrum?

Q4. Write short answers of any six parts from the following.

x6=12 2

(i) Why atomic radii cannot be determined precisely?

(ii) Define electrode potential.

(iii) Name factors affecting ionization energy.

(iv) Calculate Bond order of Helium molecule (He_2).

(v) Define enthalpy of atomization and give an example.

(vi) Define heat and give its units.

(vii) Differentiate between galvanic and electrolytic cell.

(viii) How is copper purified by electrolysis?

(ix) Why cationic radii are smaller than its parent atom?

Q. Section - II — 8x3=24

NOTE: Answer any three questions from the following.

Q5. (a) NH_3 gas can be prepared by heating two solids NH_4Cl and $Ca(OH)_2$. the mixture containing 100g of each. Calculate no. of grams of NH_3 produced.

4

(b) Define and explain Hydrogen bondings by giving any two suitable examples.

4

Q6. (a) Define plasma and explain its four applications.

4

(b) Explain the concept of orientation of orbitals by using magnetic quantum number.

4

Q7. (a) How ionization energy varies in periodic table?

4

(b) What is internal energy? Discuss first law of thermodynamics.

4

Q8. (a) $N_2(g)$ and $H_2(g)$ combine to give $NH_3(g)$. The value of Kc in this reaction at 500°C is 6.0×10^{-2}. Calculate the value of Kp for this reaction.

4

(b) Explain half life method for measurement of the order of a reaction can help us to measure the order of even those reactions which have fractional order.

4

Q9. (a) Explain elevation of boiling point with a graph.

4

(b) Explain electrolysis of aqueous solution of salts.

4